# INFORMATION PROCESSING AND RETRIEVAL

INSTITUTO SUPERIOR TÉCNICO 2025/26

# PROJECT v0

Consider the IR tasks of **keyword extraction** and **text summarization**. For extracting keywords from a document, some IR systems rank the relevance of normalized noun phrases in a document. For summarizing text, IR systems commonly follow *abstractive* or *extractive* approaches. While abstractive approaches are grounded on generative language models, extractive approaches select the most informative segments from the original document, producing a summary where these segments are ordered by relevance or location in the original document. The fundamental difficulty of these tasks lie in identifying and ranking relevant text that capture the main ideas behind a document, while ensuring minimum redundancy.

## Material

*BBC News Summary* repository with the document collection and the corresponding summaries.

**Document collection**. The corpus is composed of 2225 journalistic texts collected from the online newspaper BBC News, $D = \{d_1, .., d_{2225}\}$, along five categories: business, entertainment, politics, sports and tech news, $D = D_{business} \cup D_{entertainment} \cup D_{politics} \cup D_{sports} \cup D_{tech}$.

| **category** | **size** | **document** (avg #tokens/text) | **summary** (avg #tokens/text) |
|---|---|---|---|
| business | 510 | 328.88 ± 135.8 | 139.93 ± 59.4 |
| entertainment | 386 | 330.62 ± 261.5 | 144.05 ± 124.0 |
| politics | 417 | 453.97 ± 299.8 | 195.71 ± 139.6 |
| sports | 511 | 329.26 ± 187.8 | 143.19 ± 80.8 |
| tech | 401 | 502.70 ± 239.6 | 213.84 ± 111.4 |

Table 1: BBC News Summary: essential statistics

**Reference summaries**. The corpus contains the corresponding summaries, $R = \{r_1, .., r_{2225}\}$, also termed extracts, obtained using a state-of-the-art extractive method[1]. There is no need to be familiar with this method for the current project. Note that reference extracts may not correspond to the optimal summaries, although they offer a reference ground for summarization.

**Download**. *BBC News* repository is temporarily made available at: `https://fenix.tecnico.ulisboa.pt/downloadFile/3096843219121707/BBC%20News%20Summary.zip` (4.3Mb)

**Acknowledgments**. BBC (`http://mlg.ucd.ie/datasets/bbc.html`), Kaggle (`https://www.kaggle.com/datasets/pariza/bbc-news-summary`), D. Greene, P. Cunningham, Pariza Sharif

[1] Kernel text subspace clustering method that ensures both centrality and non-redundancy: *Greene and Cunningham. Practical Solutions to the Problem of Diagonal Dominance in Kernel Document Clustering, ICML 2006.*

## Project goals

The target IR system will be developed in two steps:

1. in the first delivery, keyword extraction and summarization will be done unsupervisedly over the $D$ collection. Reference extracts are uniquely used to guide evaluation;
2. in the second delivery, we will explore the structure of the corpus and use the reference extracts as feedback to guide the summarization process.

# Project I (*first delivery*)

**Topics**: IR models, IR evaluation, indexing, ranking, text processing, prompting

**Task**: On the first part of the project, students should be able to establish a sound solution for the keyword extraction and summarization tasks using *classic IR querying* principles.

**Grading**: **50%** sound IR system + **50%** question answering

## Functionality

Program the following functions:

1. indexing($D$,args)

| | |
|---|---|
| @input | collection $D$ and optional arguments on text preprocessing and IR models |
| @behavior | process $D$ to build document- or/and collection-level hybrid indexes with inverted (lexical search) and vector (semantic search) components |
| @output | hybrid index(es) $I$, indexing time, and in-memory size |

2. summarization($d$,$p$,$l$,$I$,args)

| | |
|---|---|
| @input | document $d$, max number of sentences ($p$) and/or characters ($l$), index $I$, and optional arguments on the approach (e.g. extractive *vs* abstractive) |
| @behavior | summarizes $d$ against $I$ according to args, $p$ and $l$ |
| @output | text if abstractive or ordered pairs (sentence position in $d$, score) if extractive |

3. keyword_extraction($d$,$p$,$I$,args)

| | |
|---|---|
| @input | document $d$, max number of keywords $p$, index $I$, optional args |
| @behavior | extracts the top informative $p$ keywords in $d$ against $I$ according to args |
| @output | ordered set of $p$ keywords |

4. evaluation($S_{set}$,$R_{set}$,args)

| | |
|---|---|
| @input | set of summaries $S_{set}$ produced from $D_{set} \subseteq D$ (e.g. a single document, a category of documents, the whole collection), the corresponding reference extracts $R_{set}$, and optional arguments (e.g. evaluation metrics, IR models) |
| @behavior | assesses the produced summaries against the reference ones |
| @output | quality of extractive summaries (e.g. F-measuring at predefined $p$-or-$l$, AUPRC) and generative summaries (e.g. embedding-mediated views) |

## Guiding principles

### Extractive design options

Multiple solutions can be envisioned to answer the summarization and keyword extraction tasks. Term-and-document frequencies can be assessed at **sentence**, **document**, or **collection** levels. Your choices will affect the index and behaviors of (2)-(3). As such choices are core to the delivery, your faculty hosts will be unable to offer definite clues on best solutions. Get creative and, when pursuing multiple solutions, compare their efficacy. Otherwise, disclose the rationale.

### IR models

Under extractive stances, *keyword extraction* should be primarily grounded on **TF-IDF** criteria, while *summarization* should further support **BM25** scoring and **embeddings** produced by pre-trained **encoder LLMs**. BM25 can be considered with the default parameters, $k_1$=1.2 and $b$=0.75, not forgetting that the average length parameter should coherently correspond to sentence- or document-level estimates depending on the selected strategy. Test two lightweight encoder LLMs with distinct profiles (e.g. different architectures, learning setups, pretraining tasks).

### Text processing

Your vector spaces should not only include words but also noun phrases. Fix a single and sound text processing pipeline, motivating its choice. Assessing the impact of text processing options, such as lemmatization *vs* stemming, is considered to be out of the scope of the project.

### Handling redundancy

A notable problem with selecting keywords or sentences with high informational value is the potential redundancy among them. In the context of summarization, the old paper "*The use of MMR: diversity-based reranking for reordering documents and producing summaries*" by Carbonell and Goldstein proposes Maximal Marginal Relevance (MMR) to increase sentence diversity. The proposed method iteratively selects the most informative sentence, adding it to the set of sentences in the summary and removing it from the document. Next sentence selection is based on the MMR score that simultaneously attempts to select sentences that are relevant to the revised document and dissimilar to the sentences selected so far (non-redundancy),

$$\mathrm{MMR}(s) = (1-\lambda) \times sim(s,d) - \lambda \sum_{v \in S} sim(s,v), \tag{1}$$

where $S$ is the sentences composing the summary at a given time and $sim$ is the cosine similarity.

We ask you to augment your summarization system with MMR. Fixing one IR model and $\lambda$=0.5, assess the impact of MMR against a redundancy-unaware summarization system on the collection $D$. Then, picking a random document, further assess the impact of varying $\lambda$.

### Generative design options

Prompt pretained decoder LLMs to perform summarization and keyword extraction, comparing their solutions against extractive methods. To this end, design two distinct prompts, providing the necessary instructions for conformative outputs, and input them to two lightweight **autoregressive LLMs** from HUGGINGFACE, ensuring your choices yield distinct profiles (e.g. different architectures, learning setups or pretraining corpora; multi-purpose Q&A *vs* summarizers).

### Ensembling

The diversity of approaches (sparse *vs* dense representations, extractive *vs* abstractive stances) can lead to arbitrarily-high differences on the produced keywords or summaries. In the presence of keyword or sentence ranks, Reciprocal Rank Fusion (RRF) can be used to place consensus,

$$\mathrm{RRF}(s) = \sum_{f \in F} \frac{1}{\mu + \mathrm{rank}(f(s))},$$

where $F$ is the compared approaches (e.g. extractive summaries with BM25 *vs* BERT, keywords under extractive *vs* generative approaches), $\mu$ is a constant that can be fixed at $\mu$=5, $s$ is the ranked object (e.g. sentence or keyword), and $rank(f(s))$ is the rank of $s$ using approach $f$.

We ask you to assess the RRF-based consensus impact on the *extractive summarization* task only. In particular, whether *consensus* from sparse and dense text representations aids behavior.

### IR system evaluation

Keywords can be manually inspected for a few selected documents, and the agreement between extractive and generative approaches briefly assessed.

Sound performance evaluation is only necessary for the summarization task.

The behavior of the extractive summarization system should be primarily assessed based on: i) precision-recall curves and average uninterpolated precision for non-fixed summary lengths, and ii) $F_\beta$-measure for pre-fixed summary lengths (where $\beta$ should be fixed in accordance with the perceived importance of precision *vs* recall).

The quality assessment of abstractive summarizations should be based on neural-based representations of both the generated and reference (or extractive) summaries using embeddings produced by an independent lightweight LLM encoder.

Evaluation can be conducted for a specific selected document, category, or overall collection. For the later cases, multiple estimates are acquired, thus particular attention should be placed to measure behavioral variability. Herein, statistical comparisons can be further undertaken to assess the differences in summarization capacity across approaches and news' categories.

## Questions to explore in the report

Guidance on angles to conduct your analyzes and write your report:

1. Are keywords extracted under TF-IDF criteria and decoder LLMs in agreement? Why?
2. How does the extractive summarization system perform for the full collection? And within each category? Any intuition for the observed differences?
3. How IR models affect summaries? Are dense representations more effective than sparse ones? Which LLM encoder achieves the highest performance? Why?
4. Considering MMR, how $\lambda$ impacts the accuracy (against reference extracts) of summaries? Should $\lambda$ be a fixed threshold or depend on the provided topic document ($d$-specific)?
5. Under embedding-based quality stances, is abstractive summarization superior to extractive summarization? Why?
6. Is Reciprocal Rank Fusion (RRF) useful to aid decisions?

# Project II (*second delivery*)

**Topics**: supervised and unsupervised guidance

**Task**: In this second stage of the IR system development, we will consider two major strategies:

- assess the added value of the relevance feedback acquired from reference extracts
- assess whether clustering can aid keyword extraction and summarization

**Grading**: **2/3** relevance feedback + **1/3** unsupervised guidance

## A) Supervised IR using reference summaries

In the presence of relevance feedback from reference extracts, summarization can be aided:

1. by **fine-tuning pretrained decoder LLMs**. Select a lightweight autoregressive LLM well-prepared for summarization, fine-tune its behavior using a training set of reference extracts, and assess performance effects on a testing set using text similarity scores such as ROUGE;
2. by reformulating summarization as a **supervised learning-to-rank** task. Herein, sentences from a document can be represented using descriptive features (e.g., positional information, sentence-to-document similarity, ranking scores) to learn a mapping between sentence features and its inclusion in a reference summary.

   This classification task should be trained using a dedicate set of training documents and the corresponding extracts. Students should select two distinct classifiers (e.g. random forests, neural networks, XGBoost, kNN, naïve Bayes) with their hyperparameterization being optional. Note that your classification-based summaries should rank sentences, not simply define inclusion *vs* exclusion. To assess the classification support, compare summarization performance in the absence and presence of feedback, ensuring that the selected testing documents do not belong to the set of documents used to trained the summarization model.

### Functionalities to implement

1. fine-tuning($M$,$D_{train}$,$R_{train}$,args)

   @input pretrained model $M$, training documents $D_{train}$ and reference extracts $R_{train}$, optional arguments on the fine-tuning process
   @output fine-tuned model $M'$

2. feature_extraction($s$,$d$,$I$,args)

   @input sentence $s$, the enclosed document $d$, optional index $I$ and arguments
   @output sentence-specific feature vector

3. training($D_{train}$,$R_{train}$,args)

   @input training docs $D_{train}$ and extracts $R_{train}$, args on the classification process
   @output classification model $M$

4. summarization($d$,$M$,$p$,$l$,args)

   @input document $d \in D_{test}$, classifier $M$, summary limits ($p$ and/or $l$), guiding args
   @output summary $s$ of document $d$

5. supervised_evaluation($D_{test}$,$R_{test}$,$M$,args)

   @input testing docs $D_{test}$ and reference extracts $R_{test}$, classifier $M$, guiding args
   @output confusion-based scores (e.g. precision, recall, AUC) of classifier $M$

6. evaluation($S_{set} \subseteq S_{test}$,$R_{set} \subseteq R_{test}$,args) – from Project I, expanded with ROUGE scores

**Questions to explore**

1. Does fine tuning help aligning summaries with reference expectations?
   According to ROUGE, how the fine-tuning setup (e.g. number of epochs) impacts performance?
2. Does the incorporation of feedback from reference extracts significantly impact the performance of the classification-based IR system? Hypothesize why is that so.
3. Which features appear to be more relevant to the target summarization task?
   Are the learned classifiers able to generalize from one category to another? Justify

## B) Unsupervised IR

To aid extractive approaches in keyword extraction and summarization, the MMR component can be replaced by a clustering-based approach to ensure the diversity (non-redundancy) of keywords and sentences. Herein, you are asked to design and implement a clustering-based approach to these ends. First, select proper word/text representations and place adequate clustering choices for selecting prototype keywords or sentences from candidate clusters. Second, assess the differences in summarization efficacy against the baseline IR system developed in the first delivery.

**Functionality**

1. sentence_clustering($d$,$I$,args)

   @input document $d$, optional index $I$ and clustering args
   @output clustering solution $C$

2. summarization($d$,$C$,$I$,args)

   @input document $d$, sentence clusters $C$, optional index $I$ and guiding args
   @output extractive summary

3. keyword_extraction($d$,$p$,$I$,args)

   @input document $d$, optional number of keywords $p$, index $I$ and args
   @output ordered set of keywords

4. evaluation($S_{set}$,$R_{set}$,args) – from Project I

**Questions to explore**

1. Do clustering-guided summarization alters the behavior and efficacy of the IR system? Hypothesize why is this so.
2. How representations and clustering choices impact summarization?
3. Is there redundancy among the extracted keywords? Suggestions for future work?

## General guidelines

- **Deadlines** for project deliveries available at the course's webpage;
- **Grading** of project deliveries: 50% first delivery + 50% second delivery;
- Please always **consult the FAQ** on the course's webpage *before posting questions to your faculty hosts*. The subject of e-mails to your faculty should be preceded by [PRI project].
- **Delivery**: report (PDF), source code and Jupyter notebook demo highlighting major facilities;
- The **templates** for the reports and notebooks are listed in the course's webpage;
- We suggest **reports** to be organized in two parts: i) major decisions placed along the proposed methodology; and ii) point-by-point direct answering of the posed questions. Answers must be supported by quantitive empirical evidence;
- **Page limits** per report are 6 main pages and 1 additional page for supplementary results;
- Students should disclose the contribution of each member at the start of the project reports whenever efforts are unequal. *Marks are individual*;
- **Submissions**: Gxx.ZIP file via Fenix, where xx corresponds to your group number. The ZIP file should contain three files: Gxx_code.ZIP with your source code, Gxx_notebook.ipynb with your notebook demo, and Gxx_report.pdf with your report;
- Please note that it is possible to submit files several times on Fenix. Yet, only the *last* submission is stored in Fenix. You can submit versions ahead to prevent late-time problems.

### Copy and plagiarism

- The project code and reports will be subjected to strict copy and plagiarism checkers;
- **Checkers** are run on both report and code against all projects (signaling LLM-mediated content) and **online** content (e.g. text, code, notebooks). Mark LLM-aided programs on your code (using the mark !!) and any followed sources (whenever applicable) for disputes;
- LLMs for programming assistance are allowed if clearly marked, yet note that evaluation is primarily focused on the soundness and critical rigor of the placed decisions;
- If copy is detected after manual clearance for any of the above cases, the registration in PRI is nullified and IST guidelines will apply. These guidelines are also valid for students sharing the copied contents, irrespectively of the underlying intent.

### Complementary notes

- The target IR system should be developed in python. Students are free to use any available python libraries to program the IR system, including SKLEARN, NLTK, TRANSFORMERS;
- Students can implement advanced summarization solutions in Kaggle or scientific literature given their proper referencing. Yet, as the project is solely evaluated on the ability to answer the requirements and questions in this statement, such direction is discouraged;
- Students can use external resources (including dictionaries, thesauri, ontologies, etc.) to guide text processing and scoring as long as their use is clearly identified.

**END**